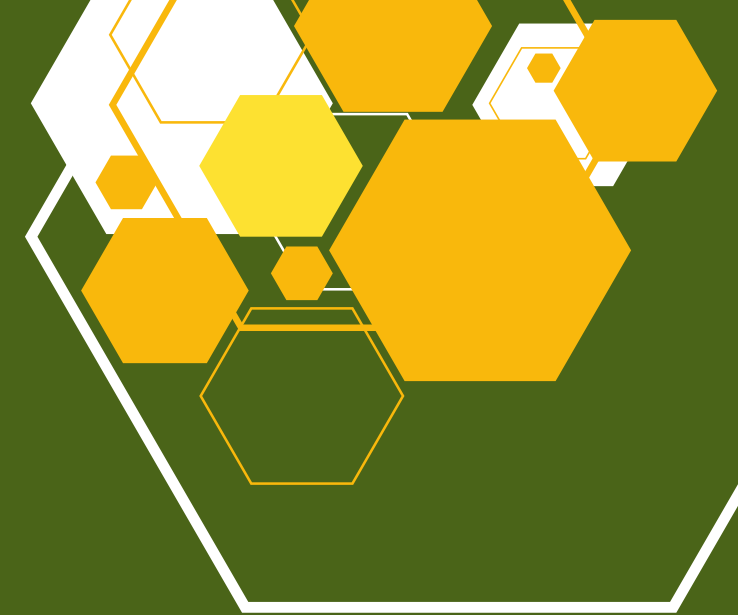
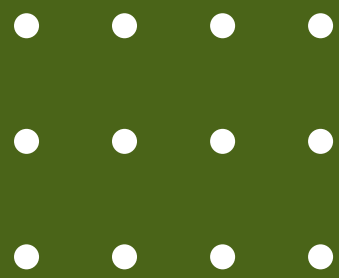
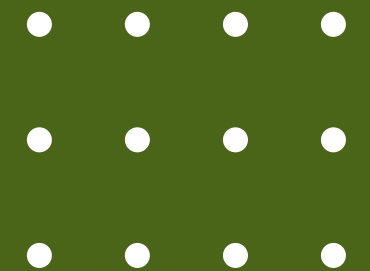
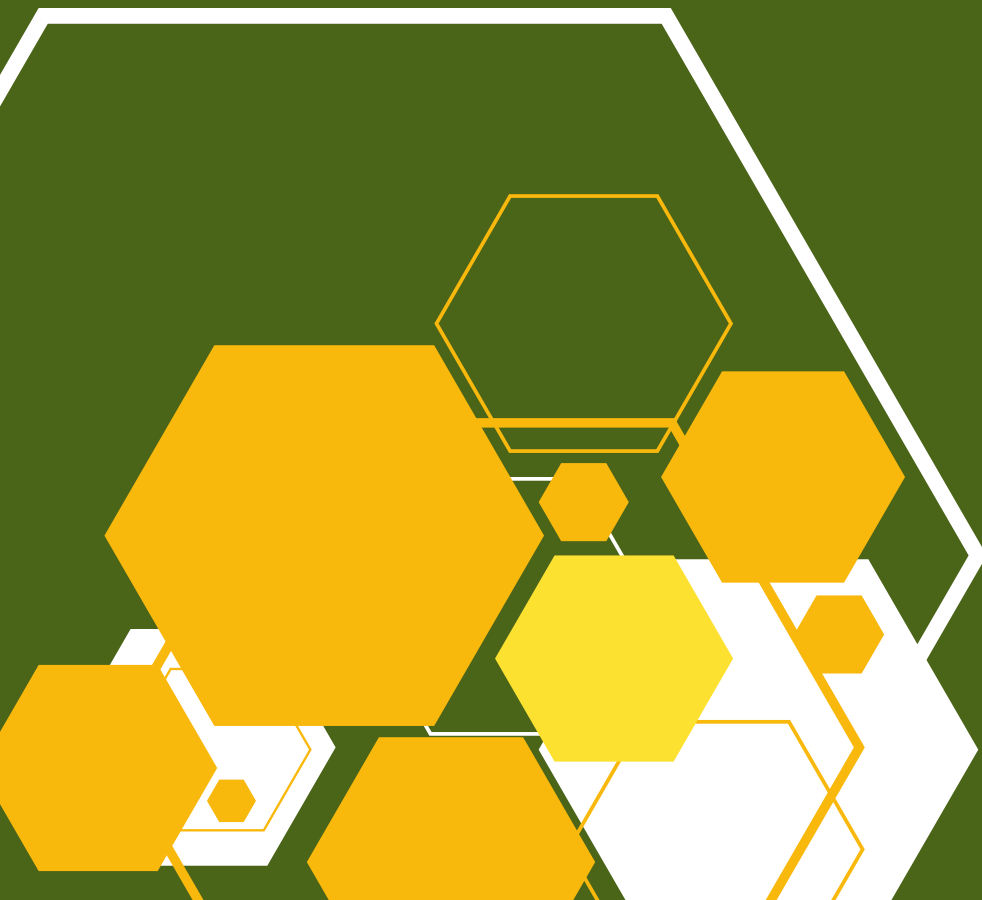




AP Statistics

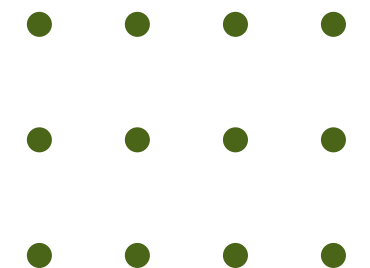
Isabel Aponte, Micaela
Velez, Mia Billinski

Instructor: Kristina Zogović
Doral Academy, Miami, FL
Semester 2025-2026



Why This Matters

- **Teen anxiety affects sleep, focus, and academic performance.**
- **CDC recommends 60 minutes of physical activity daily.**
- **Many students at Doral Academy may not meet this guidance due to pressure coming from academics or extracurriculars.**



Research Question & Hypotheses

Is there a statistically
significant negative linear
relationship between weekly
physical activity and anxiety
level?

Hypotheses:

$$H_o : \beta = 0$$

$$H_a : \beta < 0$$



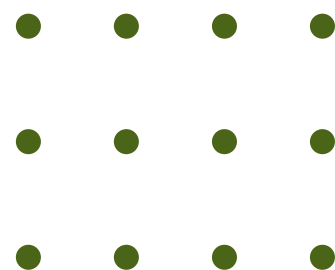
Population Comparison

Study performed to observe impact of physical activity on depression/anxiety of adolescents in Sechuan, China

[Study Link](#)

- A study conducted by a group of researchers (Zhang, Shao, Wu) from November 2024-January 2025 that consisted of a survey filled out by 6368 Chinese adolescents
- Results consisted of the correlation (if any) between depression/anxiety and physical activity
 - Our specific area of interest: correlation between physical activity (PA) and anxiety

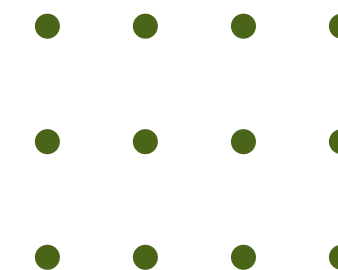
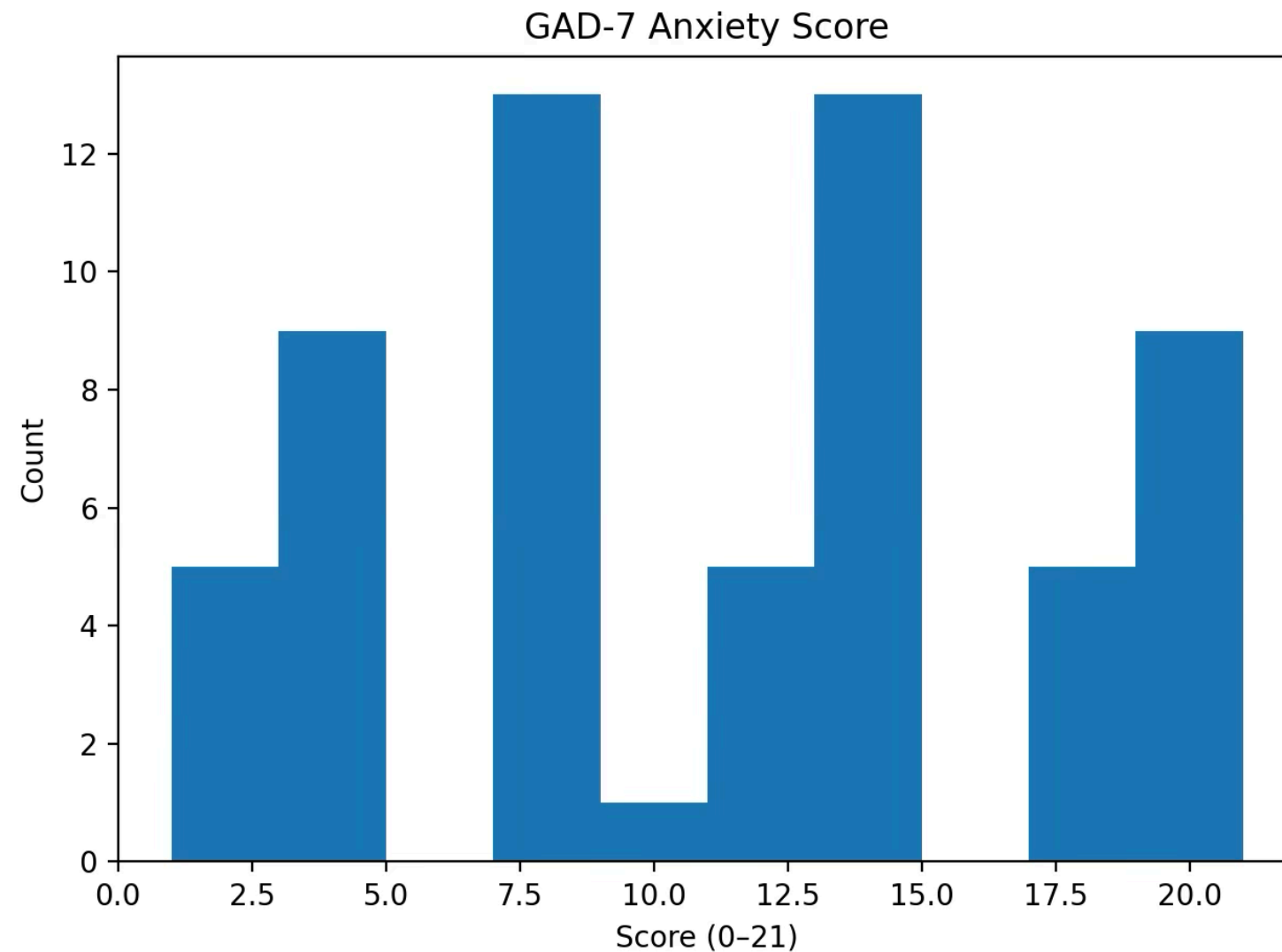
- This study concluded that, “Physical exercise was significantly negatively correlated with...anxiety...” due to the extremely low p-value of below 0.001





Sampling & Data Collection

- **Stratified Random Sample**
- **15** students per grade (afterschool sampling method)
- **n = 60** students @ Doral Academy
- It's an anonymous survey
- GAD-7 questionnaire ranging from a **0-21** scale



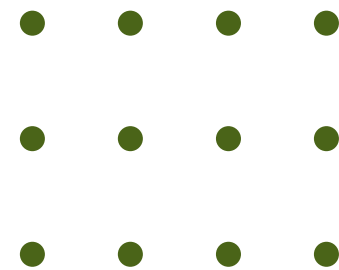
Unit 1: Descriptive Statistics

Minutes PA:

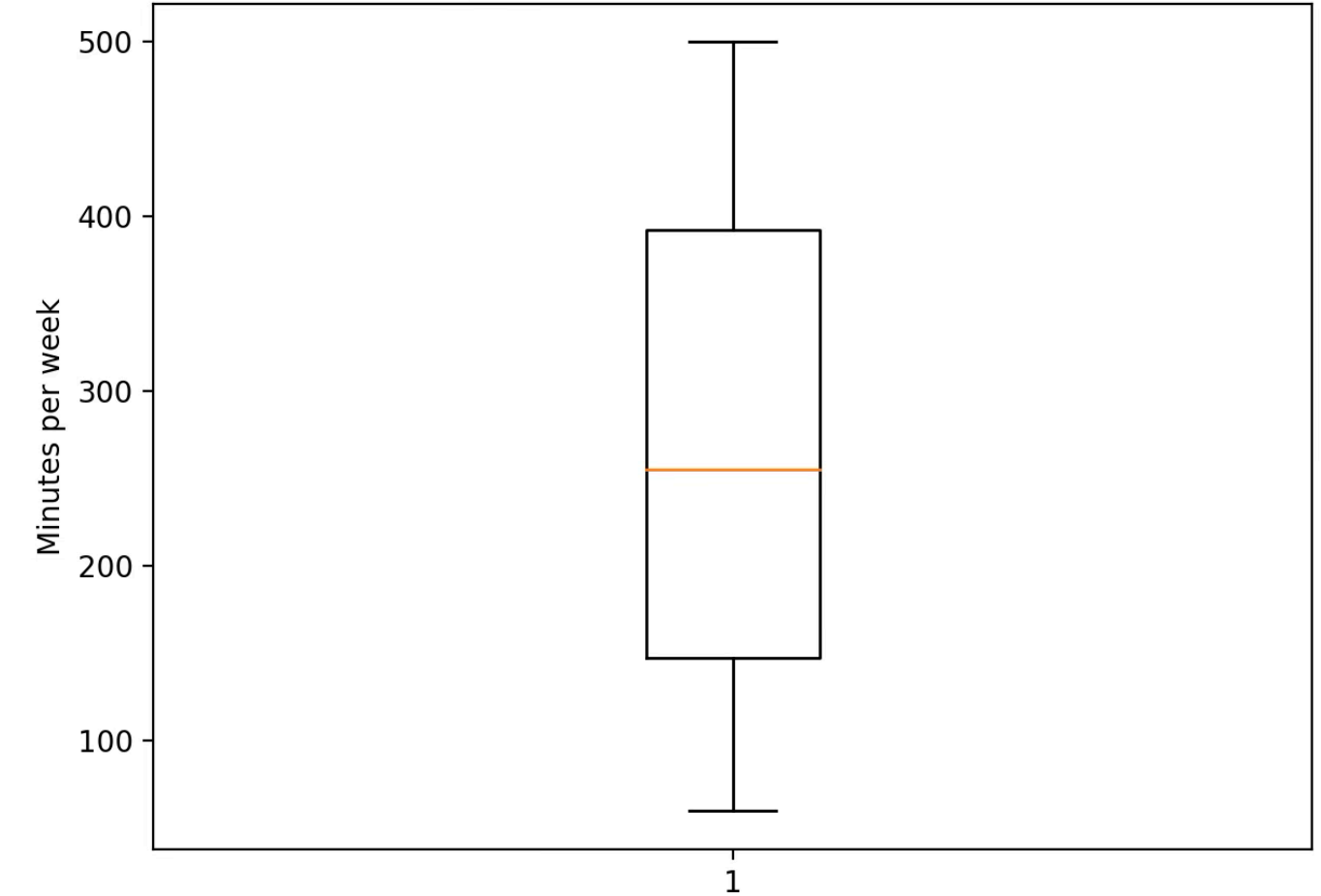
- **Mean** = 268.08
- **SD** = 140.61

Anxiety:

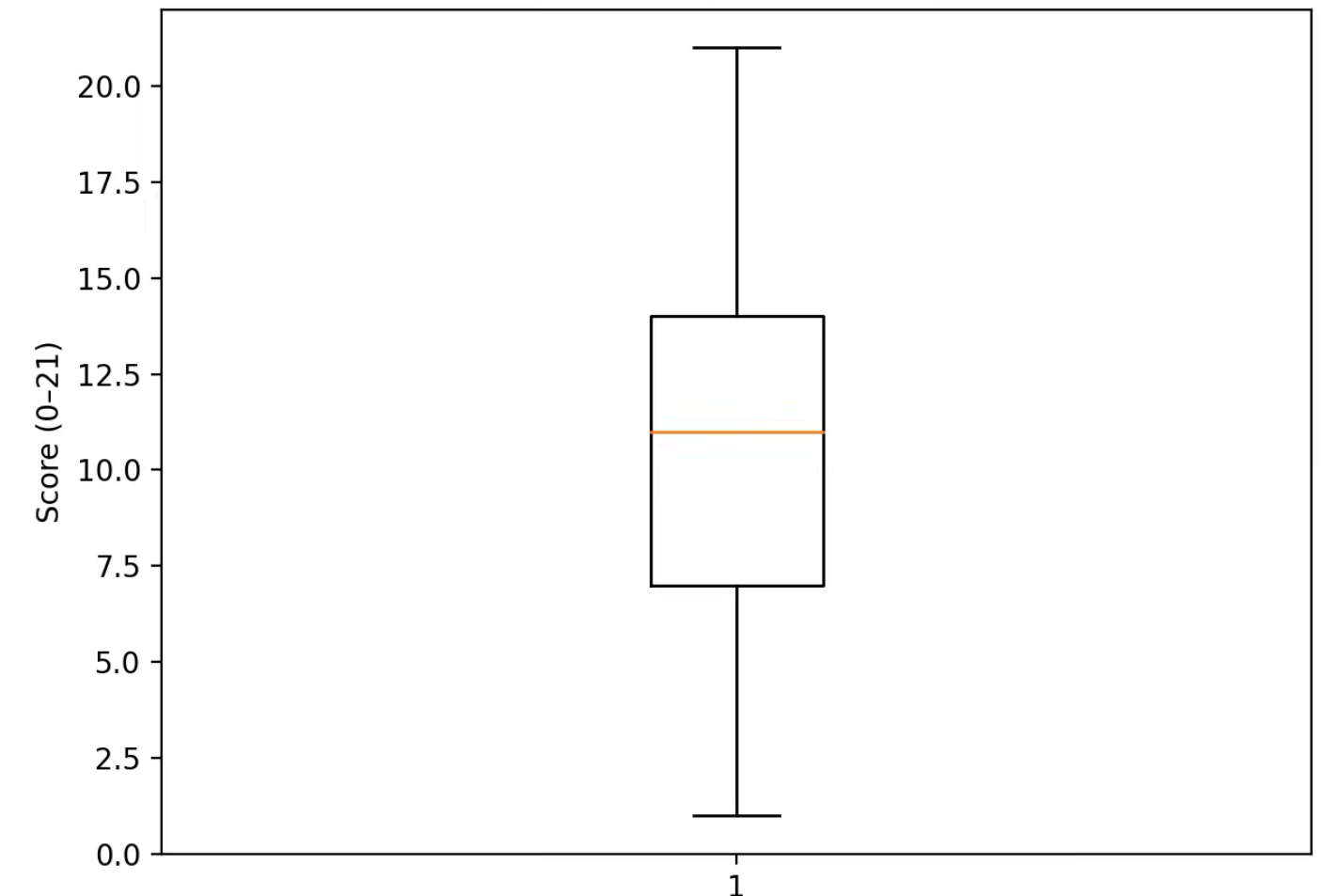
- **Mean** = 10.58
- **SD** = 6.24



Weekly Physical Activity (Minutes) - Boxplot



GAD-7 Anxiety Score - Boxplot



Unit 1-2

Our descriptive statistics

Overall descriptive statistics (Units 1–2):

Variable	Mean	SD	Five-number summary (min, Q1, med, Q3, max)
MinutesPA	268.08	140.61	(60, 147.5, 255.0, 392.5, 500)
AnxietyScore GAD-7)	10.58	6.24	(1, 7.0, 11.0, 14.0, 21)

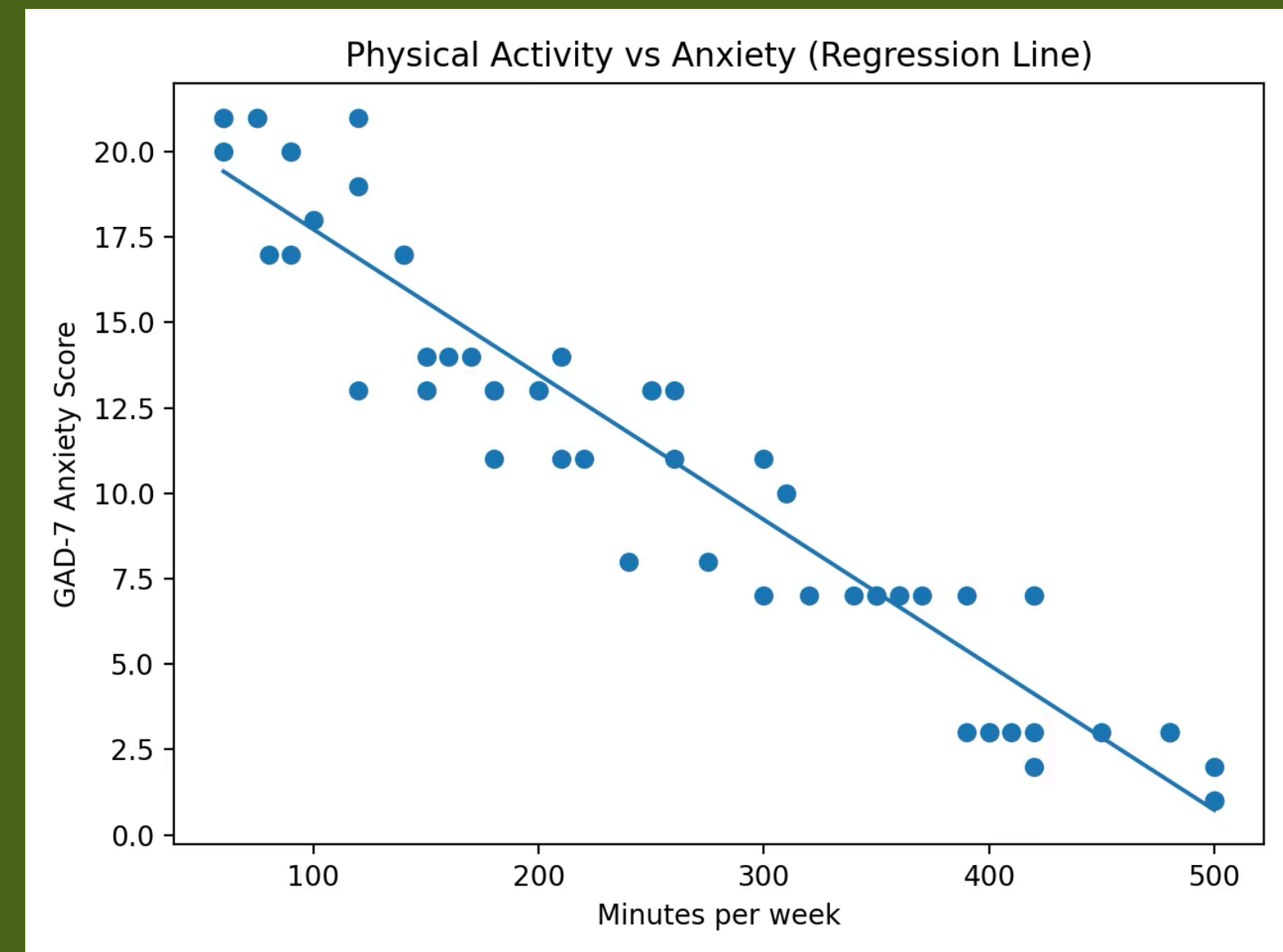
Unit 2: Scatterplot & Correlation

- **Regression Equation:**
 $\hat{y} = 21.973 - 0.04249x$
- **$r = -0.958$**

This correlation coefficient indicates that the line regression as an extremely strong negative linear association between time spent doing physical activity and anxiety. A correlation of this magnitude would be very unlikely to occur by random chance if no relationship existed.

For every additional 100 minutes of weekly physical activity, anxiety decreases by about 4.25 points

When a student does 0 minutes of physical activity weekly, the predicted anxiety level is 21.973.



Unit 6-7: Regression & Inference

Because the p- value is less than 0.0001, which is far below the significance level of 0.05, we reject the null hypothesis.

Therefore, there is enough significant statistical evidence that there is a negative linear relationship between weekly physical activity and anxiety level.

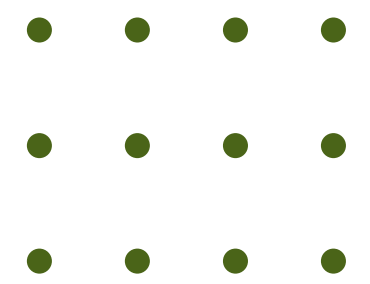
Test Results:

- $t = -25.43$
- $df = 59$
- $p < 0.0001$

The extremely large value of the t statistic comes from the very strong negative correlation observed between physical activity and anxiety, providing overwhelming statistical evidence of a linear relationship.

95% Confidence Interval:

- $(-0.04583, -0.03914)$



Calculations

Confidence Interval

$$0.1167 \pm 1.96 \sqrt{\frac{0.1167(0.8833)}{60}}$$

$$0.1167 \pm 0.0813$$
$$(0.035, 0.198)$$

One proportion z test

$$\hat{p} = \frac{7}{60} = 0.1167$$

$$p_0 = 0.246$$

$$\text{normcdf}(-99, -2.33, 0, 1) \approx$$

$$p = 0.010$$

$$0.010 < 0.05$$

↓
reject H_0

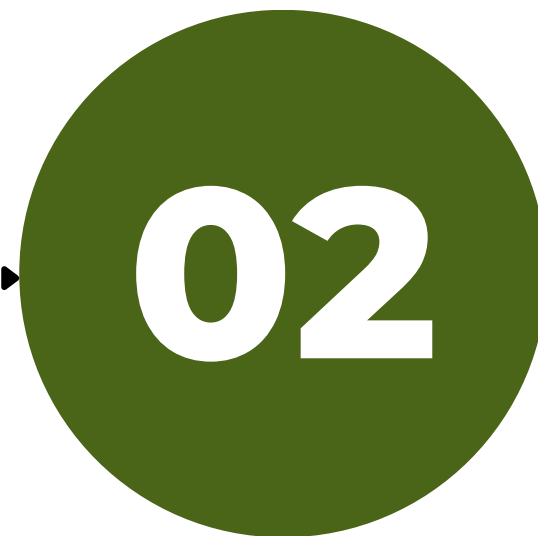
$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

$$Z = \frac{0.1167 - 0.246}{\sqrt{\frac{0.246(1-0.246)}{60}}} \approx -2.33$$

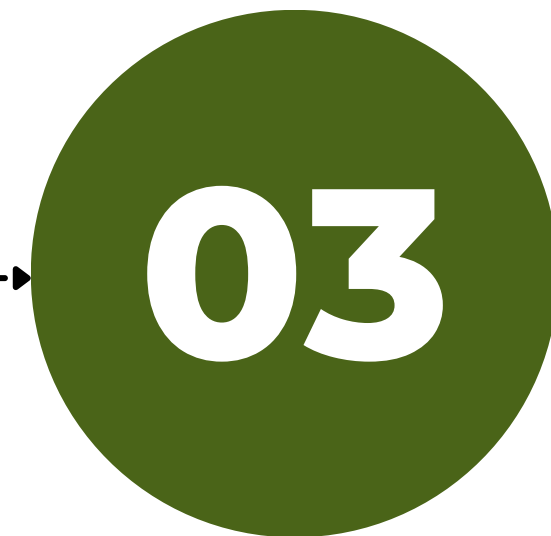
Limitations



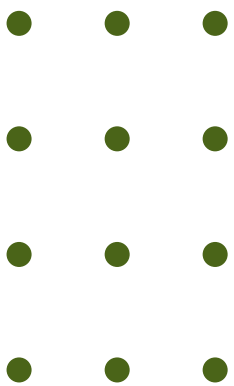
Limited to Doral Academy



It is an observational survey, and thus the findings are only in favor of association and not causation.

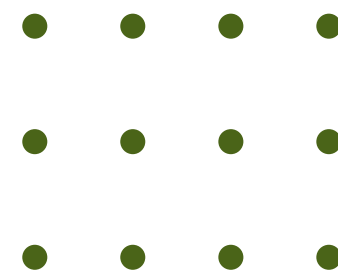
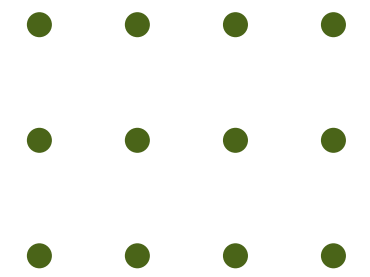


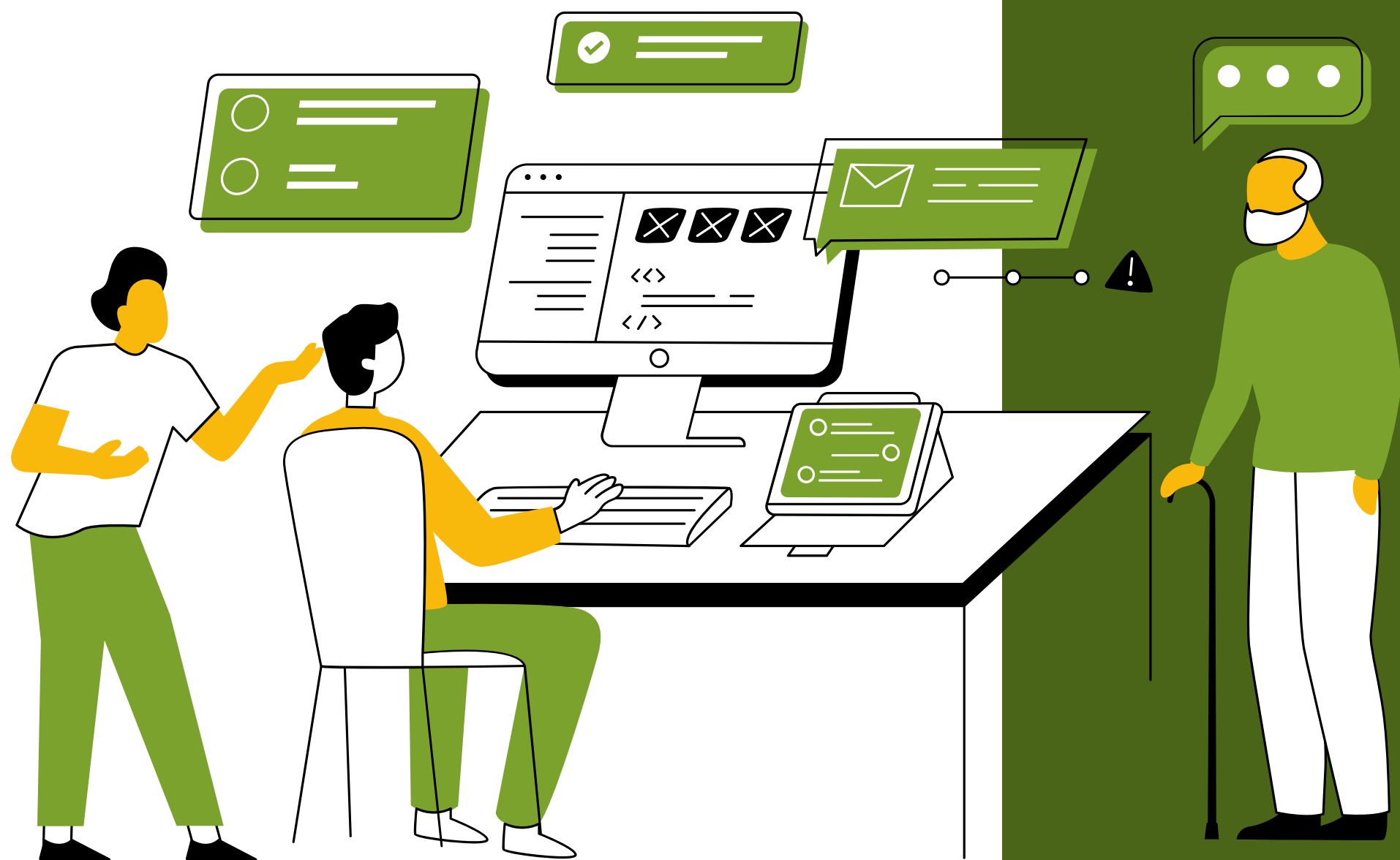
Possible Confounders: sleep, screen time, and homework may impact activity and anxiety.



Conclusion

There is enough convincing statistical evidence of a strong inverse relationship between physical activity and anxiety among Doral Academy students.





THANK YOU

